

Product Carbon Footprint Study Report

Product Name: tablet production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: tablet production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 Item(s) tablet production

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 Item(s) tablet production.

2. System Boundary

consumer electronics production, mobile device, tablet, in China.

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of tablet production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **82.237 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

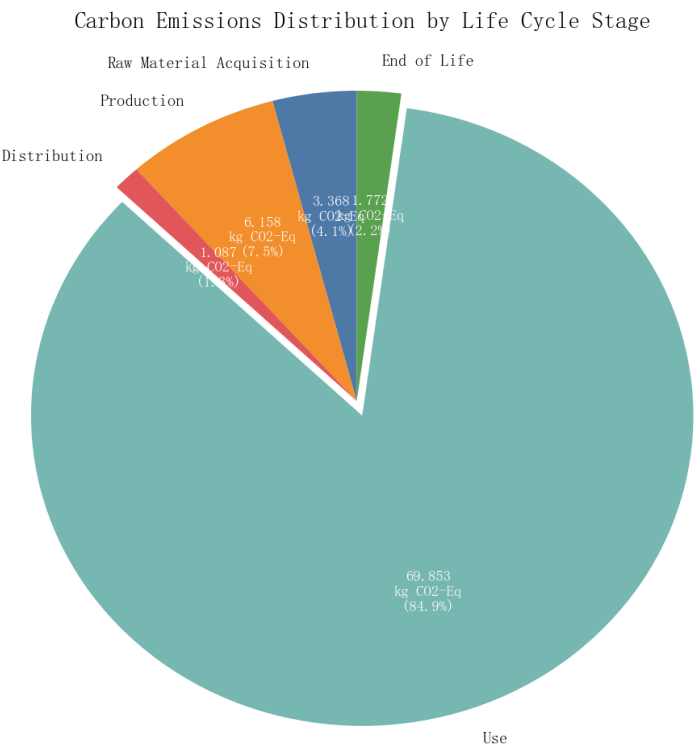


Table 1: Carbon Emissions by Life Cycle Stage

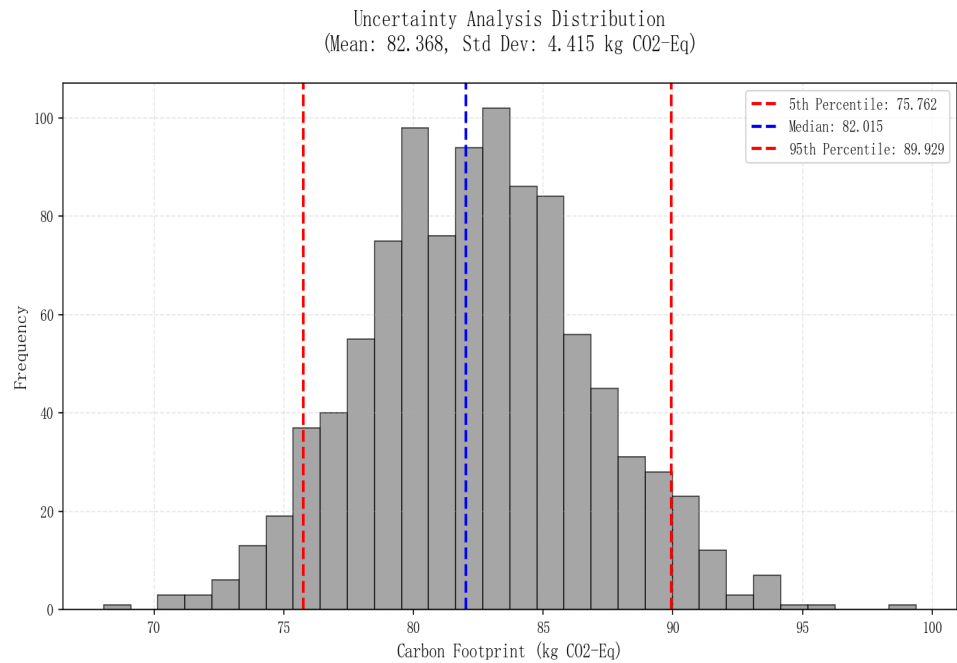
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	3.368	4.10
Production	6.158	7.49
Distribution	1.087	1.32

Use	69.853	84.94
End of Life	1.772	2.15
Total	82.237	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	82.368
Standard Deviation	4.415
Coefficient of Variation (CV)	5.36%
Median	82.015
5th Percentile	75.762
95th Percentile	89.929

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the tablet production is calculated at 82.237 kg CO₂-Eq, which represents a moderate environmental impact for a consumer electronic device. This value is primarily driven by the use stage, contributing 69.853 kg CO₂-Eq (84.9% of the total), indicating that the majority of greenhouse gas emissions occur during the product's operational lifetime. The production stage contributes 6.158 kg CO₂-Eq (7.5%), while raw material extraction adds 3.368 kg CO₂-Eq (4.1%). Distribution (1.3%) and end-of-life (2.2%) have minimal impacts.

The dominance of the use stage suggests that electricity consumption during tablet operation is the single most significant emission source. This is typical for electronic devices with extended usage periods and moderate energy demands. In contrast, the raw material and production stages, though smaller, still represent opportunities for improvement through material selection and manufacturing efficiency. The distribution and end-of-life stages are negligible, reflecting efficient logistics and low-emission disposal or recycling processes in the assumed scenario.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a standardized usage pattern—such as 4 hours of daily operation over a 4-year lifespan—and a grid electricity mix with an average carbon intensity representative of the target market. It is also assumed that the tablet is charged and used under typical conditions, and that end-of-life treatment follows a common recycling or landfill route. These assumptions significantly influence the use-stage contribution and should be validated with real-world data.

Limitations arise from the unknown top process contributions, which indicate that the underlying life cycle inventory may lack granularity or that specific unit processes were not disaggregated. This reduces the ability to pinpoint exact emission hotspots within each stage. Data quality is considered moderate, as the uncertainty analysis shows a coefficient of variation of 5.36% and a 90% confidence interval of 75.762–89.929 kg CO₂-Eq, suggesting acceptable but not high precision. The system boundary is cradle-to-grave, covering raw material extraction, production, distribution, use, and end-of-life, but excludes upstream capital goods and infrastructure. This boundary restriction may underestimate total impacts but is standard for product-level LCAs.

3. Suggestions for Improvement

- ****Improve energy efficiency during the use stage:**** Implement advanced power management features, such as dynamic voltage scaling and low-power display modes, to reduce electricity consumption. Encourage users to enable energy-saving settings and extend battery life through software updates.
- ****Optimize manufacturing processes:**** Substitute high-emission raw materials (e.g., virgin plastics) with recycled or bio-based alternatives. Increase the use of renewable energy in production facilities to lower the production-stage footprint.
- ****Promote sustainable end-of-life practices:**** Design tablets for easier disassembly and recycling, and establish take-back programs to ensure proper e-waste treatment, reducing landfill emissions.

- ****Enhance data quality:**** Disaggregate unknown top processes by obtaining supplier-specific emission factors or conducting process-level measurements. This will improve the accuracy of hotspot identification and enable targeted reduction strategies.
- ****Adopt renewable energy for charging:**** Encourage the use of grid electricity with a lower carbon intensity, or bundle the product with a renewable energy certificate, to directly reduce the dominant use-stage impact.